

## Invalidity Methods

There are at least three ways to approach Invalidity problems. The simplest, and ultimately the best, is to learn to read what the formulas say, and thus to see directly what is required to make each premise true and what is required to make the conclusion false. For example, the argument " $\exists xFx. \exists xGx \therefore \exists x(Fx \wedge Gx)$ " says that something is F, and something is G. It concludes that something is both F and G. So our universe must contain at least two individuals, an F one, and a G one; but so long as the F's and the G's don't overlap, the conclusion will be false.

A second way, discussed in the text, is to do a truth functional expansion on each formula. Initially, you may use the Expand button to do this for you, but eventually you will have to learn how to do this on your own. A truth functional expansion reduces the Chapter III Invalidity problem to a Chapter II Invalidity by Truth Tables problem. If you wish, you can even copy the expanded argument into the Truth Tables module as a User Problem (there is a button to do this). Solve the Truth Table problem by finding a row in which the (expanded) premises are all true and the (expanded) conclusion is false. The Truth Table row tells us, for each atomic formula of the form " $Gi0$ " or " $Hi1$ ", whether it is true or false. And that, in turn, tells us whether a certain individual belongs in the extension of a monadic predicate. For example, if, in the critical row in the Truth Table, we had

|       |       |
|-------|-------|
| $Gi0$ | $Hi1$ |
| F     | T     |

we would know that 0 should **not** be in the extension of the monadic predicate, "G", and that 1 **should** be in the extension of the monadic predicate "H".

A third way is to use the Derivation button, which copies the problem into the Derivation module. If, using derivational techniques including the method of EI/UI, you can cause the elimination method to terminate, and you can find an assignment of truth values to atomic formulas that makes all the quantifier-free formulas true (including those resulting from the negation of the conclusion). You can use that assignment the way it is used in the Truth Table module to construct a model.